

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	4

Case Study Topic: Google Maps Live Re-routing using Dijkstra's Algorithm on Updated Road Networks

Word Done Summary:

In this case study, Dijkstra's Algorithm was implemented to simulate the route-planning and re-routing mechanism used by navigation systems such as Google Maps.

The road network consists of eight major locations in Bangalore:

IND - Indiranagar
KOR - Koramangala
MGR - MG Road
HSR - HSR Layout
BTM - BTM Layout
JPN - Jayanagar
SRJ - Silk Board Junction
EC - Electronic City

The objective is to find the shortest route from **Koramangala (KOR)** to **Electronic City (EC)**. During the journey, an accident causes the road between **MGR and HSR** to become unavailable. The routing system must recompute the shortest path using the updated road network.

Implementation Details:

1. Construction of a weighted graph.
2. Application of Dijkstra's Algorithm.
3. Priority Queue operations.
4. Route recomputation after road closure.
5. Comparison of routes before and after the update.
6. Analysis of real-world routing systems.

Result:

Shortest Path Before Closure

KOR → HSR → BTM → EC

Distance: 21 Minutes

Shortest Path After Closure

KOR → HSR → BTM → EC

Distance: 21 Minutes

Dijkstra's Algorithm successfully computed the shortest route from Koramangala to Electronic City. Even after a major road closure, route recomputation quickly determined whether the optimal path had changed. This demonstrates how modern navigation systems adapt dynamically to traffic conditions and maintain efficient route guidance.

Output:

```
Shortest Distance = 21
Path: KOR -> HSR -> BTM -> EC
```

GitHub Link:

<https://github.com/saanvikompelli-cmd/DSA-2-Case-Studies>

Student Signature	Faculty Signature
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